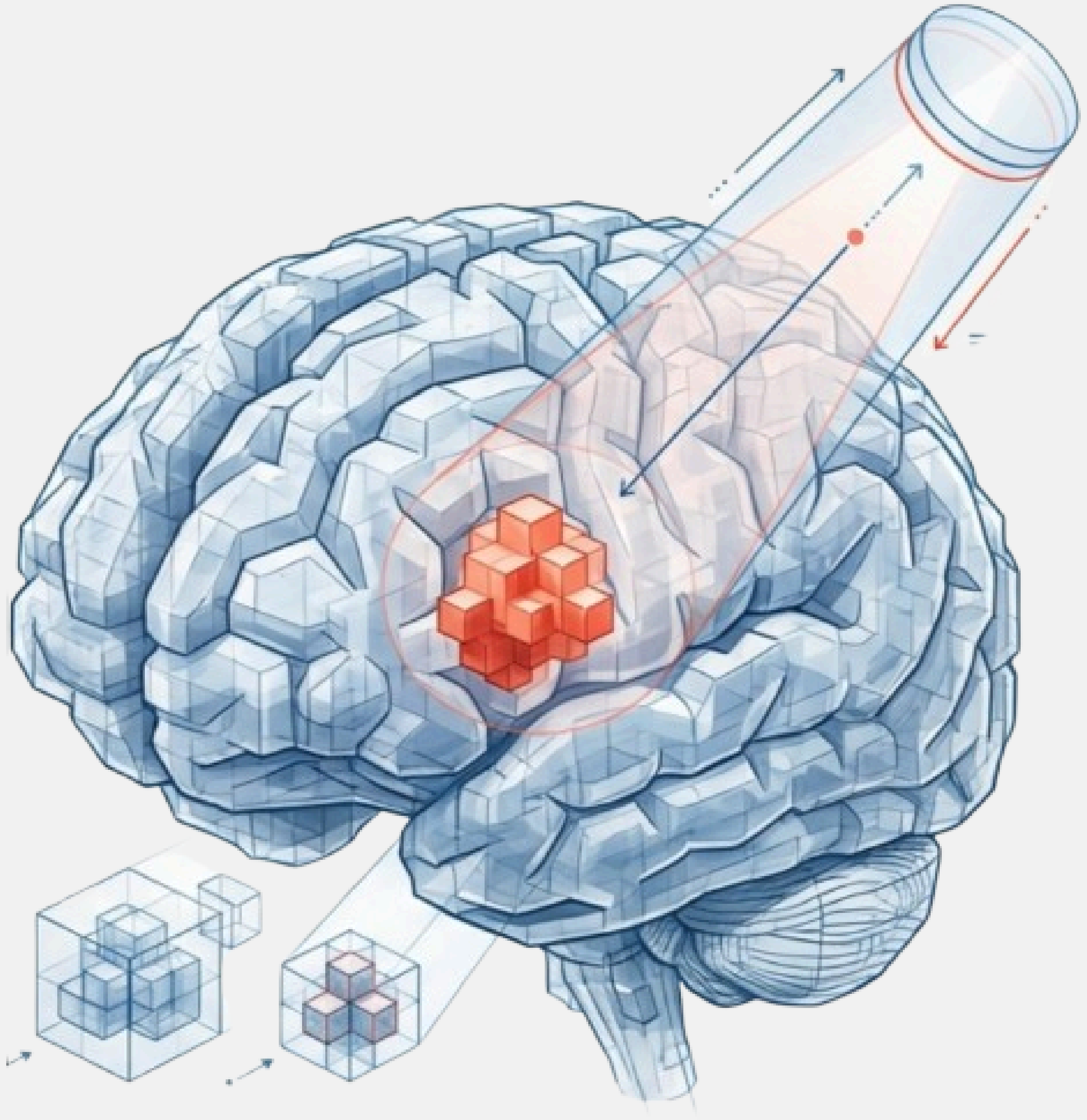
# Illuminating the Black Box

An Explainable AI framework for brain tumor segmentation using volumetric data interpretation in 3D CNNs



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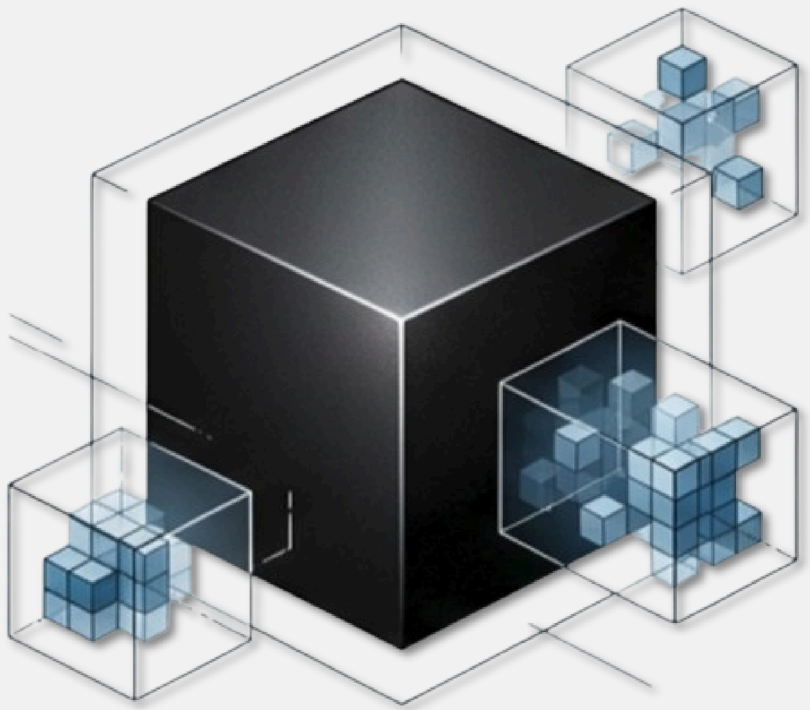
# The Trust Gap in High-stakes Neurosurgery



The Black Box

- **THE PROBLEM LANDSCAPE**

- 3D models achive high accuracy but remain opaque
- IN high-stakes neurosurgery, uncertainity is not an option
- Clinicians need the **“why”** brhind the classification

- **PROJECT AIM**

**-** Illumate this black box nature using Explainable AI (XAI)
**-** Bridge the Gap between the algoratmic accuracy and clinical trust

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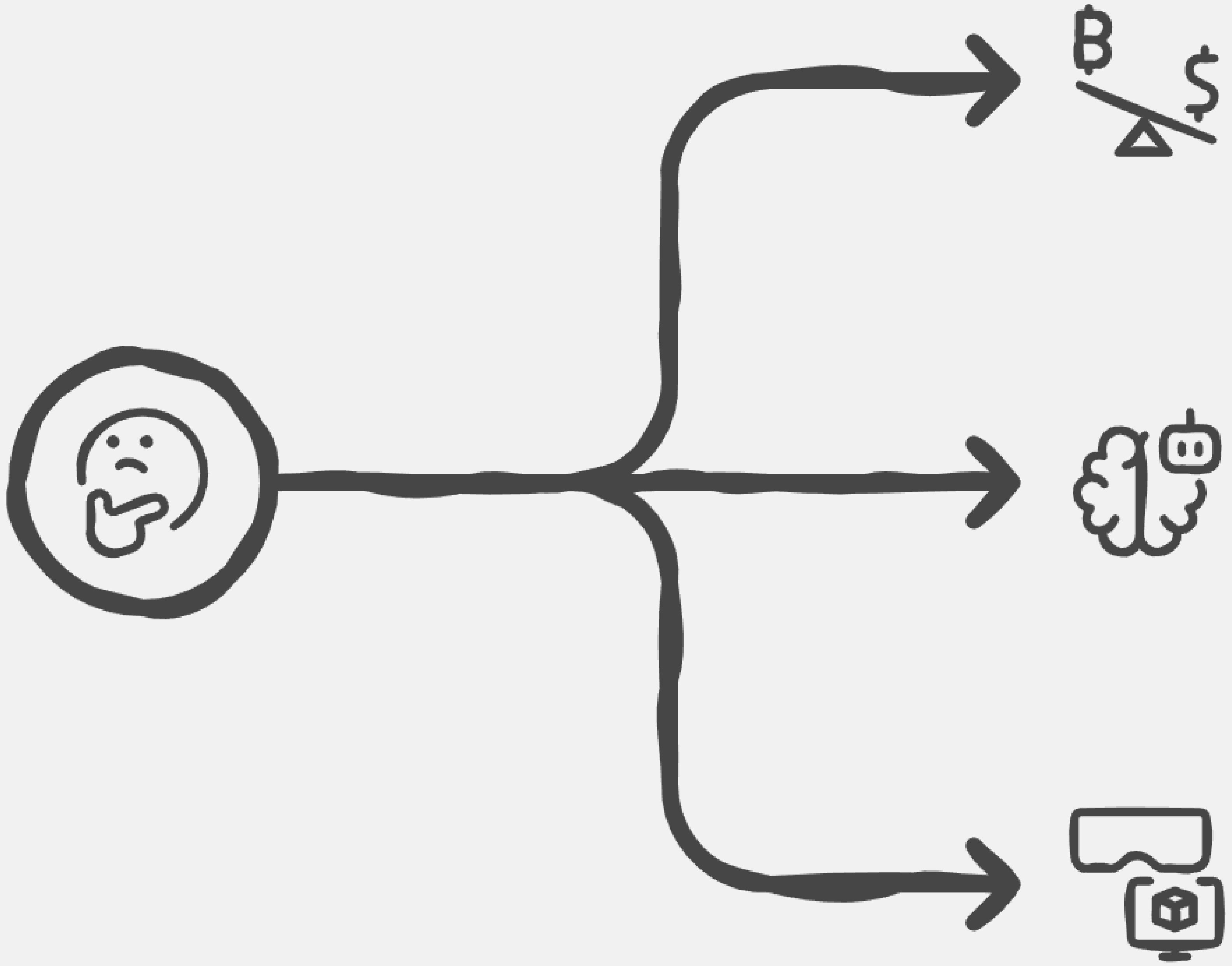
# The path to Hybid Explainability



**Quantify Trade-off**
Analyze the balance between spatial transparency and segmentation accuracy to optimize model performance.

**Implement 3D CNN with XAI**
Integrate Explainable AI techniques into a 3D CNN model to enhance understanding of segmentation results.

**Create VR Environment**
Develop a VR environment for interactive analysis, allowing for immersive exploration of segmented data.

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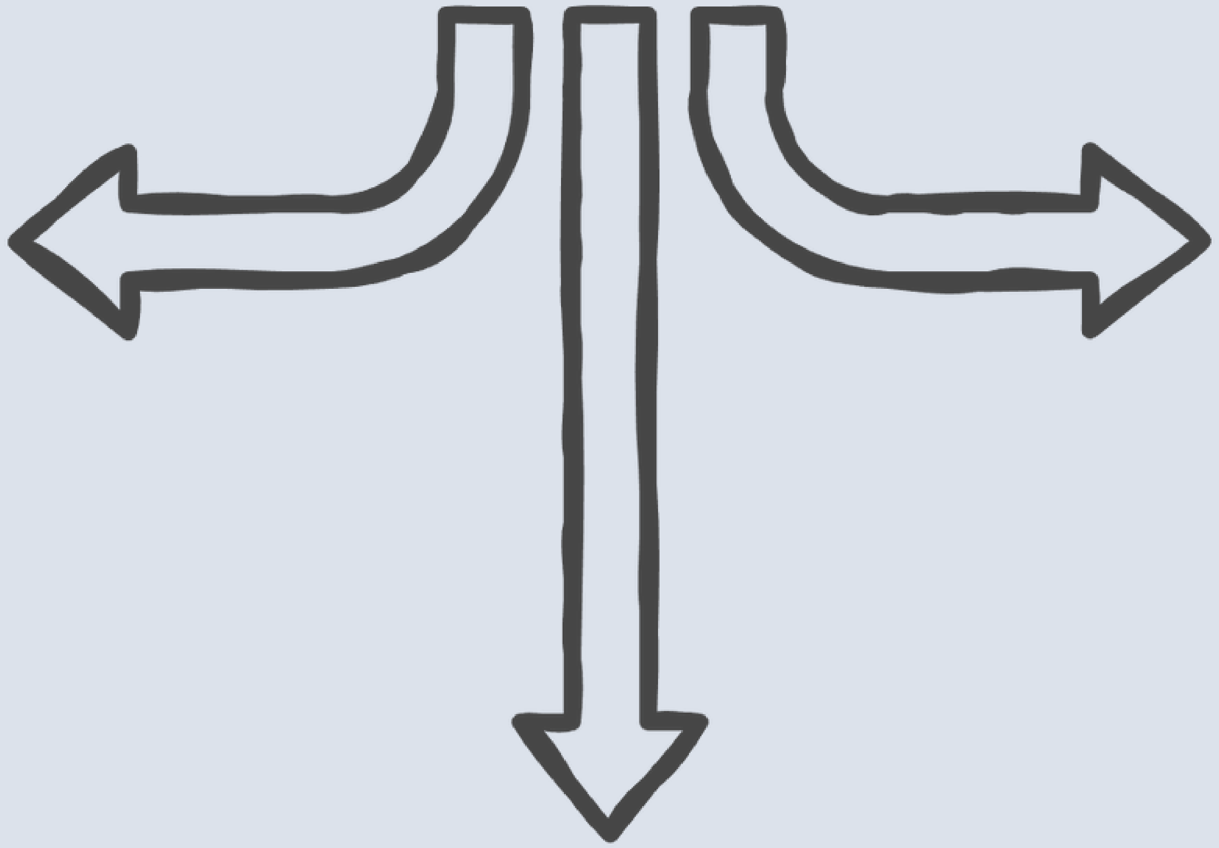
# Literature Review Achivements

1. **3D volumetric superiority:** Confirmed that 3D CNNs outperform 2D models by capturing spatial context along the Z-axis (depth).
2. Analyszed the success of the 3D CNNs , spiecifically the use of the **'dice coefficient'** to handle class imbalance( small tumor vs large background) .
3. **The Fidelity Gap:** Significant discrepencies between prediction and ground truth, hidden by model opacity.
4. **Cognitive Friction:** Increased error rates caused by analyzing 3D pathology on 2D screens.

# Evaluation Metrics

## Which evaluation metric should be used for 3D CNN segmentation?



**Dice Similarity Coefficient**
Measures overlap between ground truth and predicted area, suitable for general segmentation accuracy.

**Intersection over Union (IoU)**
Secondary measure of spatial overlap, useful for confirming accuracy.

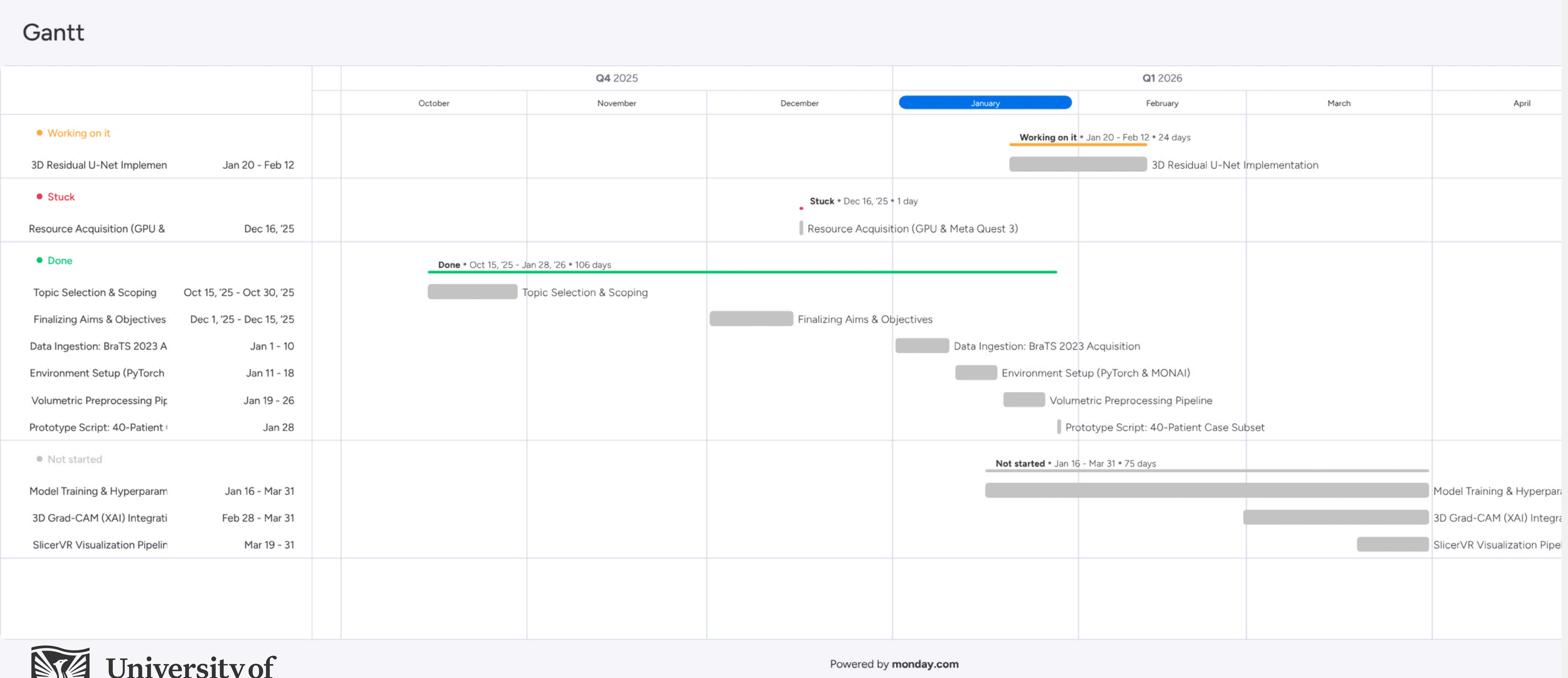
**Hausdorff Distance (HD95)**
Evaluates boundary distance, critical for surgical precision.

# Methodology & Implementation

## Current Accomplishment

- Pipeline development and project implementation planning.
- **Phase 01** : Data engineering & preprosessing
  - Volumetric **augmentation** implemented to expand dataset diversity.
  - Data cleaning and formatting is finalised.
- Working on Constructing 3D CNN model to test on preprossed data.

# Gantt Chart


Gantt
Q4 2025
Q1 2026
October
November
December
January
February
March
April
Working on it
3D Residual U-Net Implemen
Jan 20 - Feb 12
Working on it • Jan 20 - Feb 12 • 24 days
3D Residual U-Net Implementation
Stuck
Resource Acquisition (GPU &
Dec 16, '25
Stuck • Dec 16, '25 • 1 day
Resource Acquisition (GPU & Meta Quest 3)
Done
Done • Oct 15, '25 - Jan 28, '26 • 106 days
Topic Selection & Scoping
Oct 15, '25 - Oct 30, '25
Topic Selection & Scoping
Finalizing Aims & Objectives
Dec 1, '25 - Dec 15, '25
Finalizing Aims & Objectives
Data Ingestion: BraTS 2023 A
Jan 1 - 10
Data Ingestion: BraTS 2023 Acquisition
Environment Setup (PyTorch
Jan 11 - 18
Environment Setup (PyTorch & MONAI)
Volumetric Preprocessing Pip
Jan 19 - 26
Volumetric Preprocessing Pipeline
Prototype Script: 40-Patient
Jan 28
Prototype Script: 40-Patient Case Subset
Not started
Not started • Jan 16 - Mar 31 • 75 days
Model Training & Hyperparam
Jan 16 - Mar 31
Model Training & Hyperpara
3D Grad-CAM (XAI) Integrati
Feb 28 - Mar 31
3D Grad-CAM (XAI) Integra
SlicerVR Visualization Pipelin
Mar 19 - 31
SlicerVR Visualization Pipel
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# References


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# Thank You

## Q & A